School of Electrical Engineering and Telecommunications

ELEC3705 Fundamentals of Quantum Engineering

Term 3, 2026

Assignment

Due date: Friday 23 October 2026, 17:00

Assignments submitted after the due date will attract a penalty of 10% per day (including weekends).

Cutoff date: Monday 26 October 2026, 11:55

The submission portal will close on the cutoff date. No submissions will be accepted beyond that date and time.

The assignment report should be typed in a word processor of your choice. It should include electronic graphs and figures to illustrate the results of your calculations, with appropriate axis labels and figure captions. **We require that your figures are easily readable, which must be done by setting axis font sizes to 18, and plot line widths to 2.**

At your discretion, you may include in the assignment report a printout of the important parts of the Jupyter Notebook code you wrote to answer the questions. Do not include the whole code in your report (not even as an appendix) – this must be submitted in separate files.

Upload to Moodle the following files:

1) The electronic version of your assignment report, as a single pdf file;
2) The Jupyter Notebook code used to generate the answers. The lecturer should be able to press the "run" button on your Jupyter Notebook file and reproduce the results you have shown in the report. Please add clear headings and comments to the Jupyter Notebook file to indicate which question the code below it refers to i.e. if you are answering question 1_1 then the heading above the code for this question should be 'Question 1.1'.
3) Once you are finished, combine your Jupyter notebook code and pdf report into a single zipped folder with your full name and zID.

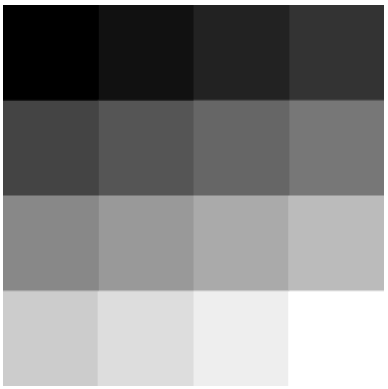
The assignment contributes a total of **20 marks** to the final course score.

PLEASE NOTE – theory and code based on this assignment will be questioned in an Oral Exam after submission. The exam will be approximately 10 minutes per student. These will be held during your allocated labs in Week 7 with allocated time slots to be released. Attendance is compulsory.

Nuclear Magnetic Resonance (NMR) [100 marks]

Nuclear magnetic resonance (NMR) is a widely employed scientific technique for investigating the structures and properties of various materials. It uses a strong time-independent magnetic field and a time-dependent oscillating magnetic field to polarize and manipulate the dynamics of nuclear spins within in the material under study.

In this question, a 2-dimensional grayscale image serves as the material of interest. The brightness of each pixel in the image corresponds to the quantity of hydrogen nuclei situated in that specific location. Figure 1 illustrates this concept using a 4x4 grayscale image as an example, where the value of each pixel represents the count of hydrogen nuclei in that specific pixel location.



0	17	34	51
68	85	102	119
136	153	170	187
204	221	238	255

Figure 1: A 4x4 grayscale image, with each pixel's value representing the count of hydrogen nuclei located in that specific pixel location.

Assuming that all hydrogen nuclei do not interact with each other within a single pixel or between pixels, the collective behaviour of spin ensembles at each pixel location can be simplified and modelled as that of an isolated hydrogen nuclear spin, with the total number of spins in that pixel location being a multiplicative factor. Similar to what has been discussed in the lecture on a single spin-1/2 electron, the Hamiltonian for a single hydrogen nuclear spin (spin-1/2) in a magnetic field is described by the following equation:

$$H = \gamma_n \, \boldsymbol{B} \cdot \boldsymbol{I} \text{ [Hz]}$$

where γ_n = 42.48 MHz/T, $\boldsymbol{B} = (B_x, B_y, B_z)$, $\boldsymbol{I} = (I_x, I_y, I_z)$. I_x, I_y, I_z are the nuclear spin matrices:

$$I_x = \frac{1}{2}\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, I_y = \frac{1}{2}\begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \text{ and } I_z = \frac{1}{2}\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

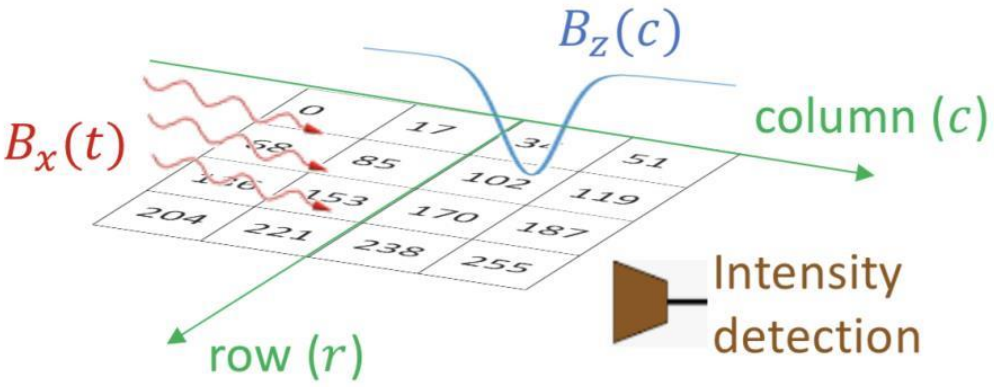


Figure 2: Static B_z field has a 1D dependency of the pixel column index, whereas B_x is a time-dependent field that does not vary based on the location. Be aware of the origin of the column and row axes.

1.1 [1 mark]

Download the task image from moodle and save it in your assignment working directory. Then use the following lines to import the image in python:

```
import matplotlib.pyplot as plt
img = plt.imread("image.jpg").astype(float)

plt.imshow(img, cmap="gray")
plt.show()
```

You should see that "img" is a 2-dimensional array which contains the values of each pixel of the given image. How many rows and columns does the image have?

1.2 [8 marks]

A strong time-independent magnetic field, denoted as **B**, is applied along the Z-direction. This field has the 1-dimensional dependency of the pixel column location c, as the following:

$$\boldsymbol{B}(\boldsymbol{c}) = B_z(c) = \left(1 - \frac{G(c)}{2maxG(c)}\right) \text{[T]} \qquad \text{Eq. 1}$$

Where G(c) is a Gaussian centred on a particular column

$$G(c) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(c-\mu)^2}{2\sigma^2}} \qquad \text{Eq. 2}$$

The column location c is measured from the middle of the image: the column with array index $n_{cols}/2$ is $c = 0$. The standard deviation, σ, characterizes the width of the distribution of the magnetic field over each column, while μ represents the central value of the magnetic field. For this assignment use σ = 10 unless otherwise specified.

With the knowledge of the magnetic B_z field, one can calculate the Zeeman energy: $E_z = \mu_n B_z$.

Assuming the magnetic B field is confined within the length of the column, plot the Zeeman energies along the column axis under two scenarios:

a) when the B_z field is centred at the middle pixel.

b) when the B_z field is centred along the column shifted 20 pixels to the right from the middle.

The x-axis of your plots should range from $-n_{cols}/2$ to $n_{cols}/2$, and the y-axis should display the values of Zeeman energies in MHz frequency units.

1.3 [6 marks]

In this and subsequent sections, unless specified otherwise, our focus is on the scenario where the B_z field is centred at the middle pixel, i.e. $B_z(64)$.

Build a Hamiltonian for each pixel in the matrix (each row is identical at this stage). What is the Hamiltonian for the central and last column?

1.4 [14 marks]

A time-dependent oscillating magnetic field, denoted as B_x, is applied across the entire image. This B_x field is uniform at each pixel location and is represented as

$$B_x(t) = B_{amp}\cos(2\pi f t)$$

where f is the frequency of the oscillating magnetic field, and B_{amp} is the amplitude of the oscillating field, which is set at $50\ mT$.

Determine the Rabi frequency of the central column and then rewrite the Hamiltonians above in the rotating frame. What is the equation of the rotating frame Hamiltonian and what is the Hamiltonian for the central and last column?

1.5 [3 marks]

As discussed in the last paragraph of the introduction, the collective behaviour of multiple spins at each individual pixel location can be effectively modelled as a single spin, multiplied by the number of spins in that pixel location. Therefore, initially, it is efficient and convenient to assume that there is only a single hydrogen nucleus at each pixel location. Be aware that even when you can consider one hydrogen nucleus per pixel, you still have multiple spins for each column because each row is composed of multiple pixels.

Using the computational basis shown below, initialize all spins in the column to spin down state.

$$|+\boldsymbol{Z}\rangle \equiv |\Uparrow\rangle = \begin{pmatrix}\mathbf{1}\\ \mathbf{0}\end{pmatrix}, \quad |-\boldsymbol{Z}\rangle \equiv |\Downarrow\rangle = \begin{pmatrix}\mathbf{0}\\ \mathbf{1}\end{pmatrix}.$$

Plot the expectation value $\langle \boldsymbol{I_z} \rangle$ for each pixel in a row. Is it what you expect?

1.6 [18 marks]

Starting with the initial state where all spins are oriented downward along the z-axis, create a time series and then simulate the time evolution of each pixel's nuclear spin under the rotating frame Hamiltonian derived in part 1.4.

Your time series should equal two periods of a full-contrast Rabi oscillations, $2T_{Rabi}$, and step in units of $T_{Rabi}/100$.

Investigate spins at the following pixel locations:

1) In the middle column
2) Offset by 5 columns from the centre
3) Offset by 10 columns from the centre
4) In the last column

Plot the expectation values of $\langle \boldsymbol{I_x} \rangle$, $\langle \boldsymbol{I_y} \rangle$ and $\langle \boldsymbol{I_z} \rangle$ against time. You should create four separate plots for the four different pixel locations. In each plot, display the expectation values using different colours to differentiate them.

1.7 [6 marks]

The $\pi/2$ pulse has the shortest pulse duration to flip the spin from $|\Downarrow\rangle$ to the state in xy-plane. What is the duration of this $\pi/2$ pulse, and does it match to what you found in part 1.6?

Plot the expectation values of $\langle \boldsymbol{I_x} \rangle$, $\langle \boldsymbol{I_y} \rangle$ and $\langle \boldsymbol{I_z} \rangle$ against pixel locations after applying the oscillating magnetic field B_x with $\pi/2$ pulse duration.

Hint: there's no need to recompute the time evolution; you can simply apply the $\pi/2$ pulse unitary operator to the initial state.

1.8 [14 marks]

Keep the static magnetic B_z unchanged but switch off the oscillating magnetic field B_x after applying the $\pi/2$ pulse. The Hamiltonian in the laboratory frame should be consistent with what you derived in part 1.3. As the result, all spins will precess at their respective Larmor frequency in the laboratory frame, starting with the initialize state determined in part 1.7.

Simulate the time evolution of each pixel's nuclear spin under the laboratory frame Hamiltonian, with the time series that spans exactly two periods of the Larmor period of the middle spin, $2T_{Larmor}$, and steps in units of $T_{Larmor}/100$.

Investigate spins in the following pixel locations:

1) In the middle column
2) Offset by 10 columns from the centre
3) In the final column

Plot the expectation values of $\langle \boldsymbol{I_x} \rangle$, $\langle \boldsymbol{I_y} \rangle$ and $\langle \boldsymbol{I_z} \rangle$ against time. You should create three separate plots for each location and use different colours to differentiate each expectation value.

1.9 [6 marks]

Up to this point, our analysis has been based on the simplification of a single spin per pixel. Now we aim to determine the aggregated signal received at the detector, which measures the total magnetic dipole of the spin ensemble along +x direction. To accomplish this, we will first multiply the expectation value of $\langle \boldsymbol{I_x} \rangle$ at each pixel by the number of spins in the corresponding pixel. Then, we will sum up these contributions from all pixels within a single row to obtain the overall intensity. Mathematically, the intensity at each row can be represented as:

$$P(r) = \sum_c N(r,c) \langle \boldsymbol{I_x} \rangle (c) \qquad \text{Eq. 3}$$

Where $\boldsymbol{N(c)}$ is the number of spins at row r and column c. Importantly this means that each row gives one intensity value. Plot the intensity of the Lamor precession over time from above (choose just one row).

1.10 [24 marks]

So far, we have only set the B_z field centred at the middle pixel in the first column of the image. To create a reconstructed 2-dimensional image, we must scan both the rows and columns to collect intensity data for each pixel within the 2D image.

To find the intensity of each column we need to change the central column of the magnetic field B_z. Which parameter in $G(c)$ do you need to change to shift the B_z field?

Reconstruct the image by determining the intensity at each pixel for the readout time corresponding to maximum intensity from part 1.9. Plot it using the code above. What is the effect of this NMR process, and does it make sense?